

Exercise 1:

a.)

```
[hawk@nixos:~]$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          48 bits physical, 48 bits virtual
Byte Order:             Little Endian
```

b.)

```
Vendor ID:              AuthenticAMD
Model name:              AMD Ryzen 7 6800H with Radeon Graphics
CPU family:              25
Model:                   68
Thread(s) per core:      2
Core(s) per socket:      8
Socket(s):                1
```

c.)

```
Virtualization:          AMD-V
Caches (sum of all):
  L1d:                    256 KiB (8 instances)
  L1i:                    256 KiB (8 instances)
  L2:                      4 MiB (8 instances)
  L3:                     16 MiB (1 instance)
NUMA:
```

d.)

```
[hawk@nixos:~]$ df -h
Filesystem                Size      Used Available Use% Mounted on
/dev/nvme1n1p2            914.8G    180.3G    688.0G   21% /
tmpfs                      7.7G         6.6M      7.6G    0% /run
devtmpfs                   1.5G          0      1.5G    0% /dev
tmpfs                      15.3G    154.2M     15.2G    1% /dev/shm
/dev/nvme1n1p2            914.8G    180.3G    688.0G   21% /nix/store
efivarfs                   128.0K     24.4K     98.6K   20% /sys/firmware/efi/efivars
none                       1.0M          0      1.0M    0% /run/credentials/systemd-journald.service
tmpfs                      15.3G     1.1M     15.3G    0% /run/wrappers
/dev/nvme1n1p1            1022.0M    167.3M    854.7M   16% /boot
tmpfs                      3.1G      4.5M      3.1G    0% /run/user/1000
df: /run/user/1000/doc: Operation not permitted
/dev/nvme0n1               915.8G    10.7G    858.5G    1% /run/media/hawk/D-drive
```

```
[hawk@nixos:~]$ cat /proc/meminfo
MemTotal:       32100272 kB
MemFree:        19346204 kB
MemAvailable:   23326728 kB
```

e.)

```
[hawk@nixos:~]$ ps -o pid,user,stat | wc -l
432

[hawk@nixos:~]$ ps -o pid,user,stat | grep "S" | wc -l
291

[hawk@nixos:~]$ ps -o pid,user,stat | grep "R" | wc -l
4

[hawk@nixos:~]$ ps -o pid,user,stat | grep "T" | wc -l
1

[hawk@nixos:~]$ ps -o pid,user,stat | grep "Z" | wc -l
7

[hawk@nixos:~]$
```

f.)

```
[hawk@nixos:~]$ grep ctxt /proc/stat
ctxt 230552999
```

Exercise 2:

Mem1

```
[hawk@nixos:~]$ cat /proc/37137/status | grep -E "VmSize|VmRSS"
VmSize:      6464 kB
VmRSS:       1832 kB
```

Mem2

```
[hawk@nixos:~]$ cat /proc/37403/status | grep -E "VmSize|VmRSS"
VmSize:      10364 kB
VmRSS:       1824 kB
```

Mem3

```
[hawk@nixos:~]$ cat /proc/37499/status | grep -E "VmSize|VmRSS"
VmSize:      6460 kB
VmRSS:       3784 kB
```

Mem4

```
[hawk@nixos:~]$ cat /proc/37589/status | grep -E "VmSize|VmRSS"
VmSize:      6464 kB
VmRSS:       5724 kB
```

Ok, so

VmSize is the memory asked for by the process.

VmRSS is the memory it gets in RAM.

Mem1 and Mem2 ask for memory for 1M and 2M size array respectively so Mem2 gets double Mem1.

Mem3 and Mem4 ask for 1M so they should be similar to Mem1.

Comparing Mem3 and Mem4 they actually access the elements so they have more RAM usage with Mem4 accessing all of the array size and Mem3 accessing half of it so we see Mem4 being roughly double that of Mem3 after removing the base line used without accessing any of the array which can be seen from Mem 1 and Mem 2.

Exercise 3:

Ok first I looked for all processes that had subprocesses in their command using `ps -ef`. Then I looked at the top most one, got it's process tree using `pstree -p <pid>` and verified which ones are it's child processes and then used `wc` to count them.

```
[hawk@nixos:/run/media/hawk/D-drive/Fox/Cs744/lab1/task3/subprocess]$ ps -ef | grep "subprocesses 0818"
42410 hawk      0:00 ./subprocesses 0818
42415 hawk      0:00 ./subprocesses 0818
42416 hawk      0:00 ./subprocesses 0818
42417 hawk      0:00 ./subprocesses 0818
42418 hawk      0:00 ./subprocesses 0818
42419 hawk      0:00 ./subprocesses 0818
42420 hawk      0:00 ./subprocesses 0818
42421 hawk      0:00 ./subprocesses 0818
42422 hawk      0:00 ./subprocesses 0818
42423 hawk      0:00 ./subprocesses 0818
42424 hawk      0:00 ./subprocesses 0818
42425 hawk      0:00 ./subprocesses 0818
42426 hawk      0:00 ./subprocesses 0818
42427 hawk      0:00 ./subprocesses 0818
42556 hawk      0:00 grep subprocesses 0818

[hawk@nixos:/run/media/hawk/D-drive/Fox/Cs744/lab1/task3/subprocess]$ pstree -p 42410
subprocesses(42410)─subprocesses(42415)
                  └─subprocesses(42416)
                     └─subprocesses(42417)
                        └─subprocesses(42418)
                           └─subprocesses(42419)
                              └─subprocesses(42420)
                                 └─subprocesses(42421)
                                    └─subprocesses(42422)
                                       └─subprocesses(42423)
                                          └─subprocesses(42424)
                                             └─subprocesses(42425)
                                                └─subprocesses(42426)
                                                   └─subprocesses(42427)

[hawk@nixos:/run/media/hawk/D-drive/Fox/Cs744/lab1/task3/subprocess]$ pstree -p 42410 | wc -l
13
```

Exercise 4:

```
[hawk@nixos:/run/media/hawk/D-drive/Fox/Cs744/lab1/task3/strace]$ strace -c ./empty
```

% time	seconds	usecs/call	calls	errors	syscall
46.59	0.000191	191	1		execve
20.24	0.000083	5	15		mmap
11.46	0.000047	5	8	6	openat
5.37	0.000022	7	3		mprotect
4.88	0.000020	4	5	4	newfstatat
1.46	0.000006	3	2		read
1.46	0.000006	3	2		pread64
1.22	0.000005	2	2		close
0.98	0.000004	4	1		munmap
0.98	0.000004	4	1	1	access
0.73	0.000003	3	1		arch_prctl
0.73	0.000003	3	1		set_tid_address
0.73	0.000003	3	1		set_robust_list
0.73	0.000003	3	1		prlimit64
0.73	0.000003	3	1		getrandom
0.73	0.000003	3	1		rseq
0.49	0.000002	2	1		fstat
0.49	0.000002	2	1		brk
100.00	0.000410	8	48	11	total

```
[hawk@nixos:/run/media/hawk/D-drive/Fox/Cs744/lab1/task3/strace]$ strace -c ./hello
```

Process ID : 47072

Enter your name : Hawk

Welcome Hawk

% time	seconds	usecs/call	calls	errors	syscall
0.00	0.000000	0	3		read
0.00	0.000000	0	5		write
0.00	0.000000	0	2		close
0.00	0.000000	0	3		fstat
0.00	0.000000	0	2	2	lseek
0.00	0.000000	0	15		mmap
0.00	0.000000	0	3		mprotect
0.00	0.000000	0	1		munmap
0.00	0.000000	0	3		brk
0.00	0.000000	0	2		pread64
0.00	0.000000	0	1	1	access
0.00	0.000000	0	1		getpid
0.00	0.000000	0	1		execve
0.00	0.000000	0	1		arch_prctl
0.00	0.000000	0	1		set_tid_address
0.00	0.000000	0	8	6	openat
0.00	0.000000	0	5	4	newfstatat
0.00	0.000000	0	1		set_robust_list
0.00	0.000000	0	1		prlimit64
0.00	0.000000	0	1		getrandom
0.00	0.000000	0	1		rseq
100.00	0.000000	0	61	13	total

Common: execve mmap openat mprotect newfstatat read pread64 close munmap
access arch_prctl set_tid_address set_robust_list prlimit64 getrandom rseq fstat brk

Unique to 2:

Write: displays something on the terminal

lseek: moves output offset

getpid: gets the process ID

Exercise 5:

```
[hawk@nixos:/run/media/hawk/D-drive/Fox/Cs744/lab1/task3/files]$ ps -ef | grep openfiles
43910 hawk      0:00 ./openfiles
43937 hawk      0:00 grep openfiles

[hawk@nixos:/run/media/hawk/D-drive/Fox/Cs744/lab1/task3/files]$ lsof -p 43910
COMMAND  PID USER  FD   TYPE DEVICE SIZE/OFF  NODE NAME
openfiles 43910 hawk   cwd    DIR   259,1    4096 47054914 /run/media/hawk/D-drive/Fox/Cs744/lab1/task3/files
openfiles 43910 hawk   rtd    DIR    0,51      500      2 /
openfiles 43910 hawk   txt    REG   259,1    8760 47054915 /run/media/hawk/D-drive/Fox/Cs744/lab1/task3/files/openfiles
openfiles 43910 hawk   mem    REG   259,3    17854919 /nix/store/571z36553l75g3l78pvxjij8z5rcsd4n-glibc-2.42-61/lib/libc.so.6 (stat: Operation not permitted)
openfiles 43910 hawk   mem    REG   259,3    17854913 /nix/store/571z36553l75g3l78pvxjij8z5rcsd4n-glibc-2.42-61/lib/ld-linux-x86-64.so.2 (stat: Operation not permitted)
openfiles 43910 hawk    0u    CHR  136,0      0t0      3 /dev/pts/0
openfiles 43910 hawk    1u    CHR  136,0      0t0      3 /dev/pts/0
openfiles 43910 hawk    2u    CHR  136,0      0t0      3 /dev/pts/0
openfiles 43910 hawk    3w    REG    0,54      0      3 /tmp/welcome to OS
openfiles 43910 hawk    4w    REG    0,54      0      4 /tmp/CS333
openfiles 43910 hawk    5w    REG    0,54      0      5 /tmp/CS347

[hawk@nixos:/run/media/hawk/D-drive/Fox/Cs744/lab1/task3/files]$
```

Got the Pid using ps -ef and then used lsof.

Exercise 6:

```
[hawk@nixos:~]$ lsblk -o NAME,FSTYPE,MOUNTPOINT
NAME          FSTYPE MOUNTPOINT
nvme1n1
├─nvme1n1p1   vfat   /boot
└─nvme1n1p2   ext4   /
nvme0n1       ext4   /run/media/hawk/D-drive
```

Block device name

File system type

Mount point